A GROUP SCHEME WHOSE IDENTITY COMPONENT IS FLAT BUT WHICH IS NOT FLAT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The monomorphism identities X x_S X = X and X x_S X x_S X = X with unique S-morphisms are the author's own carried step (3905-3908); the identity component of S amal X is the S summand.

Editorial note. The label lemma-non-flat-group-scheme names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 3890-3897: for a monomorphism of schemes X -> S, set G = S amal X and define m: G x_S G -> G on the decomposition G x_S G = (X x_S X) amal X amal X amal S by mapping the summands X x_S X and S into S and the summands X into X by the identity
- Context span 3898-3909: the group-law associativity m compose (m x id_G) = m compose (id_G x m) is verified by decomposing G x_S G x_S G into the 8 pieces, each isomorphic to S, X, X x_S X, or X x_S X x_S X; since X -> S is a monomorphism, X x_S X = X and X x_S X x_S X = X and in each case there is exactly one S-morphism S -> S or X -> X, whence the two compositions agree
- Context span 3910-3914: taking X -> S any nonflat monomorphism (e.g., a closed immersion) gives a group scheme over S whose identity component is S (hence flat) but which is not flat

Let $X \to S$ be a monomorphism of schemes. Let $G = S \amalg X$. Let $m : G \times_S G \to G$ be the S-morphism

$$G \times_S G = X \times_S X \amalg X \amalg X \amalg S \longrightarrow G = X \amalg S$$

which maps the summands $X \times_S X$ and S into S and maps the summands X into X by the identity morphism. This defines a group law. To see this we have to show that $m \circ (m \times \mathrm{id}_G) = m \circ (\mathrm{id}_G \times m)$ as maps $G \times_S G \times_S G \to G$. Decomposing $G \times_S G \times_S G$ into components as above, we see that we need to verify this for the restriction to each of the 8-pieces. Each piece is isomorphic to either S, X, $X \times_S X$, or $X \times_S X \times_S X$. Moreover, both maps map these pieces to S, X, S, X respectively. Having said this, the fact that $X \to S$ is a monomorphism implies that $X \times_S X \cong X$ and $X \times_S X \times_S X \cong X$ and that there is in each case exactly one S-morphism $S \to S$ or $X \to X$. Thus we see that $m \circ (m \times \mathrm{id}_G) = m \circ (\mathrm{id}_G \times m)$. Thus taking $X \to S$ to be any nonflat monomorphism of schemes (e.g., a closed

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immersion) we get an example of a group scheme over a base S whose identity component is S (hence flat) but which is not flat.

Lemma 0.1. *There exists a group scheme G over a base S whose identity component is flat over S but which is not flat over S.*

Proof. See discussion above. □